Pricing Derivatives Using Monte Carlo Simulation

1 Introduction

In financial mathematics, calculating the true price of derivatives, such as options, can be challenging due to the complexity of underlying asset dynamics. One powerful method for approximating option prices is the *Monte Carlo simulation*, a computational algorithm that relies on repeated random sampling to obtain numerical results. This paper explores the application of Monte Carlo simulation to approximate the prices of call options, put options, and Asian options under specific conditions, analyzing how the price and associated error vary with the number of simulations.

2 Monte Carlo Simulation for Option Pricing

The Monte Carlo simulation approximates option prices by generating a large number of possible asset price paths based on stochastic models, typically the Geometric Brownian Motion for stock prices. For each path, the option payoff is calculated, and the average discounted payoff across all paths provides an estimate of the option price. The accuracy of the approximation improves with a larger number of simulations (n), though at the cost of increased computational time and storage requirements.

In this study, we focus on pricing a *call option* using Monte Carlo simulation, with plans to extend the analysis to put options and Asian options. We examine how the number of trials (n) affects the estimated price, error, percentage error, and computational time.

3 Call Option Pricing

We consider a European call option with the following parameters:

- Initial stock price: $S_0 = 60$
- Strike price: $K = 65$
- Risk-free rate: $r = 3\%$ (0.03)
- Volatility: $\sigma = 0.3$

The Monte Carlo simulation is performed for varying numbers of trials (n), ranging from 1,000 to 1 billion. The results, including the estimated option price, error, percentage error, and time duration, are summarized in the table below. Note that simulations were stopped at $n = 1,000,000,000$ due to excessive time duration and storage requirements.

Table 1: Monte Carlo Simulation Results for Call Option Pricing

Number of Trials (n)	Price	Standard Error	Percentage Error (%)	Time Duration
1,000	5.6934272154	0.3524039067	6.1896621025404476	0.0416683750
10,000	6.0885754219	0.1170608722	1.922631552392815	0.0293462910
100,000	5.8368203180	0.0365350242	0.6259405330035073	0.0378850839
200,000	5.8966270577	0.0258306067	0.4380573244412722	0.0136703750
1,000,000	5.8979124817	0.0116255343	0.1971126951820443	0.0800332080
10,000,000	5.9015376269	0.0036732493	0.06224224188946671	0.4007488750
100,000,000	5.8987551805	0.0011606359	0.00019675945799149642	4.8253777500
1,000,000,000	5.8992701404	0.0003670859	0.0000622256	331.601870

4 Analysis

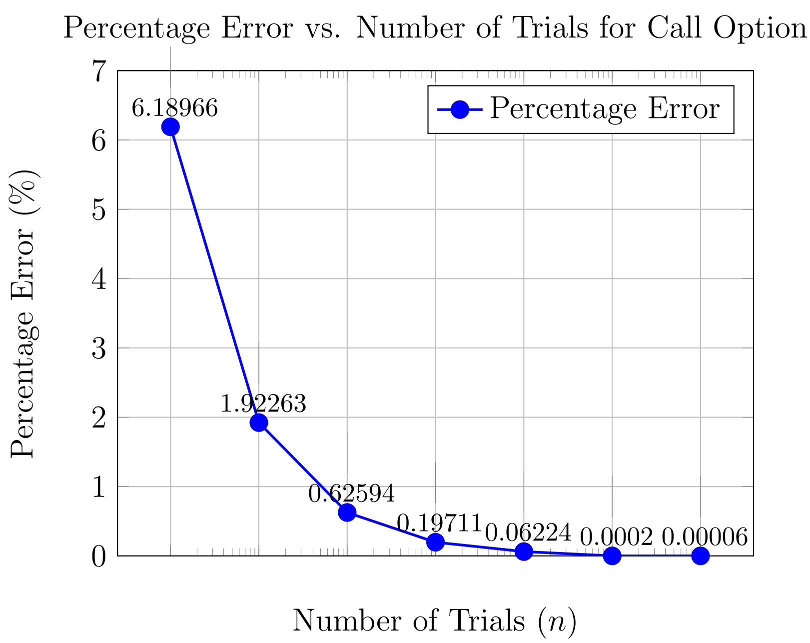


Figure 1: Relationship between the number of trials (n) and percentage error for call option pricing using Monte Carlo simulation, with labeled coordinates.

Relationship between the number of trials (n) and percentage error for call option pricing using Monte Carlo simulation, with labeled coordinates.

The simulation results indicate that at $n = 1,000,000$, the percentage error decreases to approximately 0.197%, which is well below the target threshold of 0.5%. This suggests that $n = 1,000,000$ provides a sufficiently accurate approximation for the call option price under the given parameters. Therefore, we fix $n = 1,000,000$ for further analysis to balance accuracy and computational efficiency.

Future work will extend the Monte Carlo simulation to put options and Asian options, comparing their price approximations and error behaviors under similar conditions.

5 Put Option Pricing

We consider a European Put option with the following parameters:

- Initial stock price: $S_0 = 60$
- Strike price: $K = 55$
- Risk-free rate: $r = 3\%$ (0.03)
- Volatility: $\sigma = 0.3$

The Monte Carlo simulation is performed for varying numbers of trials (n), ranging from 1,000 to 1 billion. The results, including the estimated option price, error, percentage error, and time duration, are summarized in the table below. Note that simulations were stopped at $n = 1,000,000,000$ due to excessive time duration and storage requirements.

Table 2: Monte Carlo Simulation Results for Put Option Pricing

Number of Trials (n)	Price	Standard Error	Percentage Error (%)	Time Duration
1,000	3.4977868700	0.1874162840	5.3581390440023594	0.0112356250
10,000	3.9665790303	0.0646841349	1.6307285038334316	0.0587356660
100,000	3.9429362680	0.0203820351	0.5169252994599188	0.0357832500
1,000,000	3.9458729605	0.0064305971	0.16297020139723639	0.0890570000
10,000,000	3.9467271516	0.0020326618	0.05150246566396758	0.4012099589
100,000,000	3.9505680591	0.0006432265	0.016284699418928474	5.1449937921
1,000,000,000	3.9498083425	0.0002033819	0.0000514916	350.4375698000

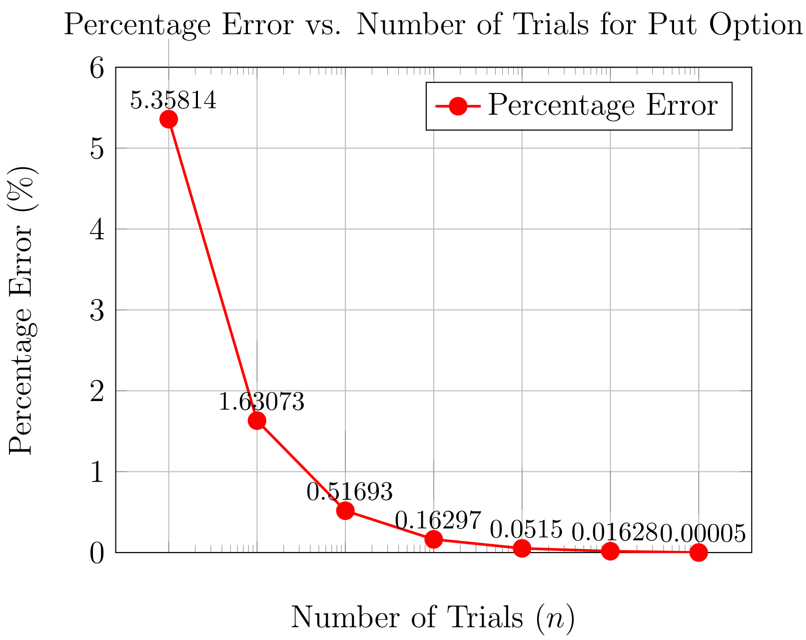


Figure 2: Relationship between the number of trials (n) and percentage error for put option pricing using Monte Carlo simulation, with labeled coordinates.

6 Asian Option Pricing

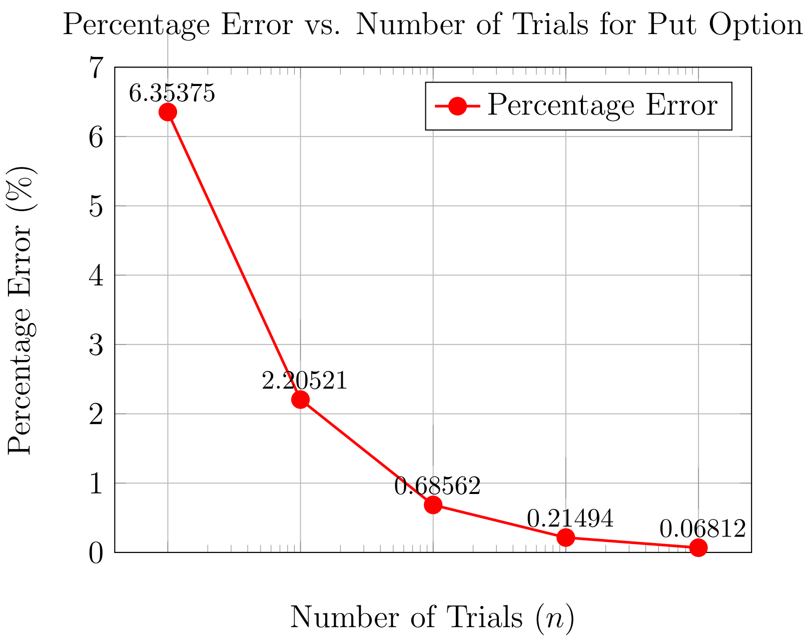
We consider a Asian option with the following parameters:

- Initial stock price: $S_0 = 60$
- Strike price: $K = 65$
- Risk-free rate: $r = 3\%$ (0.03)
- Volatility: $\sigma = 0.3$

The Monte Carlo simulation is performed for varying numbers of trials (n), ranging from 1,000 to 10 million. The results, including the estimated option price, error, percentage error, and time duration, are summarized in the table below. Note that simulations were stopped at $n = 100,000,000$ due to excessive time duration and storage requirements.

Table 3: Monte Carlo Simulation Results for Asian Option Pricing

Number of Trials (n)	Price	Standard Error	Percentage Error (%)	Time Duration
1,000	2.9493238862	0.1873925776	6.353746988893269	0.3233828751
10,000	2.4778316263	0.0546414057	2.2052106004030313	3.1388878340
100,000	2.5406059852	0.0174188498	0.6856179166712045	31.6894467090
1,000,000	2.5618290434	0.0055063970	0.21494006516673218	309.2048262920
10,000,000	2.5573364649	0.0017419616	0.06811624569485263	3344.7093135830



7 Relationship Between the Strike Price and the Percentage Error in Monte Carlo Simulation

To further investigate the behavior of the Monte Carlo simulation, we fix the number of trials at $n = 200,000$ and analyze how the percentage error of the call option price varies with changes in the strike price (K). By varying the strike price while keeping the initial stock price ($S_0 = 60$), risk-free rate ($r = 0.03$), and volatility ($\sigma = 0.3$) constant, we aim to understand the sensitivity of the pricing error to different strike price levels. This analysis will provide insights into the robustness of the Monte Carlo simulation across a range of option moneyness conditions.

Future work will extend the Monte Carlo simulation to put options and Asian options, comparing their price approximations and error behaviors under similar conditions.

Table 4: Relation Between Strike Price K and the Error for $n = 200,000$ for call option

Strike Price (K)	Option Price	Standard Error	Percentage Error (%)
65	5.9030471319	0.0260105784	0.44062969182064775
70	4.3108548296	0.0226688466	0.5258550218202654
80	2.2085308848	0.0166125454	0.7521989177289221
100	0.5045578207	0.0079014707	1.5660188782655925
120	0.1235174742	0.0041755108	3.380502102142773
145	0.0155736815	0.0013123009	8.426401459816489

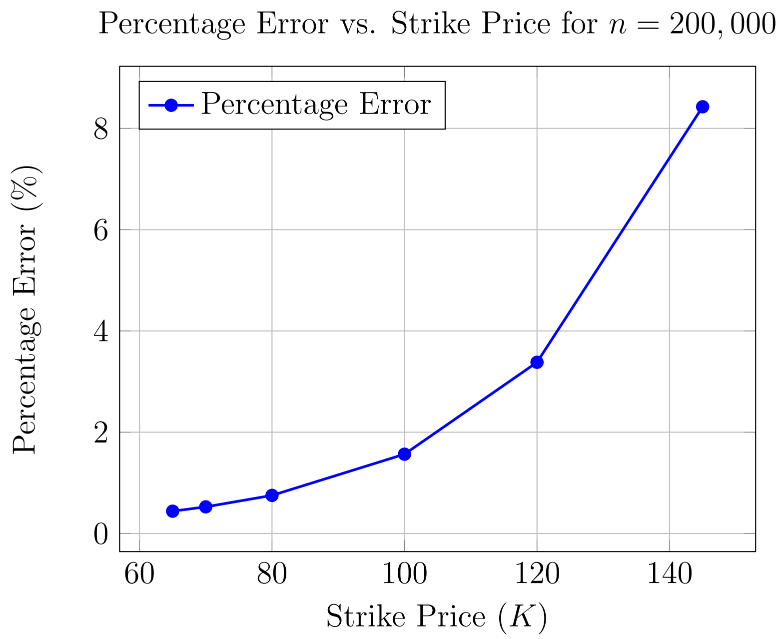


Figure 3: Relationship between strike price (K) and percentage error for Monte Carlo simulation with $n = 200,000$.

Table 5: Relation Between Strike Price K and the Error for $n = 200,000$ for put option

Strike Price (K)	Option Price	Standard Error	Percentage Error (%)
55	3.9380010143	0.0143560978	0.3645529239890197
50	2.2485225882	0.0104507612	0.464783466881606
40	0.4652423990	0.0041779326	0.8980120035938425
30	0.0347791649	0.0009082143	2.6113747848661533
25	0.0037966588	0.0002453879	6.463259689570557
20	0.0001852903	0.0000579096	31.253457061023754

Percentage Error vs. Strike Price for Put Option ($n = 200,000$)

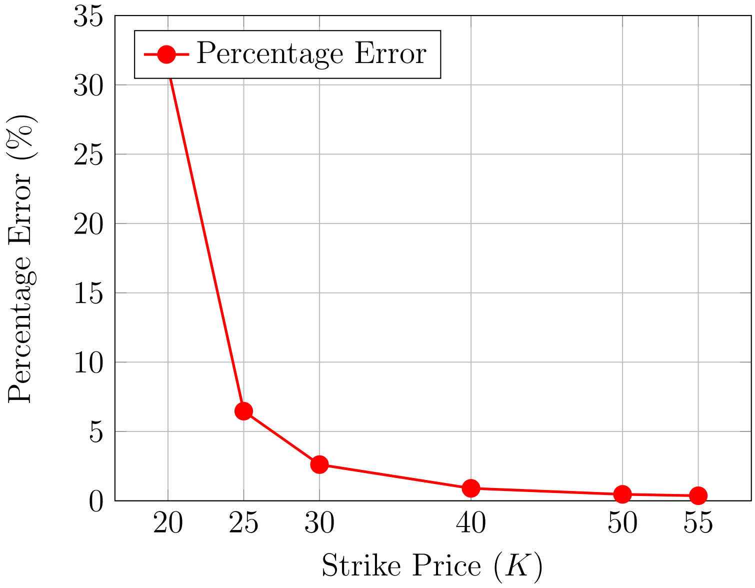


Table 6: Relation Between Strike Price K and the Error for $n = 200,000$ for Asian option

Strike Price (K)	Option Price	Standard Error	Percentage Error (%)
65	2.5574461955	0.0122824000	0.48026034648884 12
70	1.3640460181	0.0091348157	0.6696853009332701
80	0.3259984252	0.0044256591	1.3575707059465612
100	0.0137401286	0.0009628217	7.0073703249731
120	0.0006620147	0.0002653571	40.08326233750248

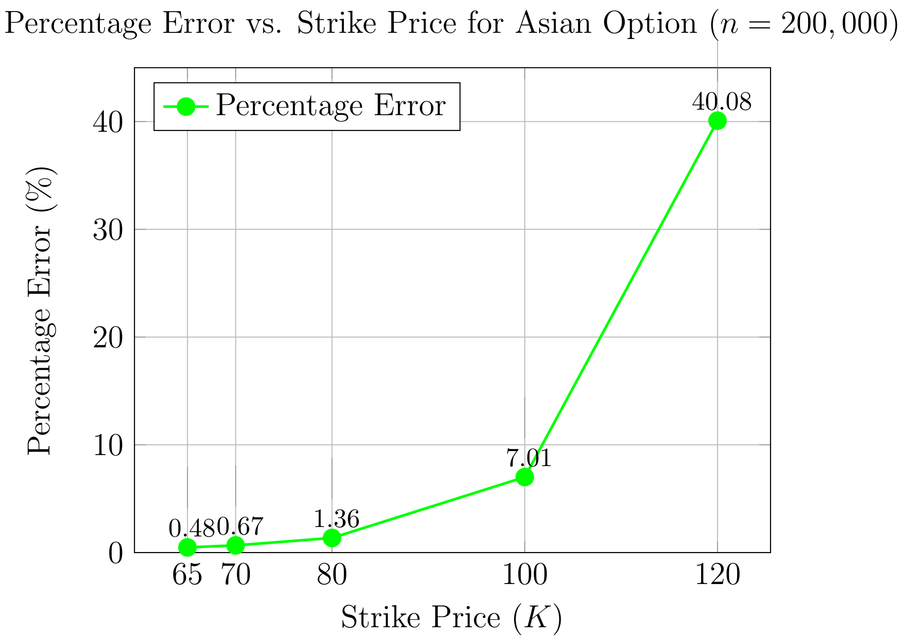


Figure 4: Relationship between strike price (K) and percentage error for Asian option pricing with $n = 200,000$, with labeled coordinates.